IF2230
Introduction to Operating Systems

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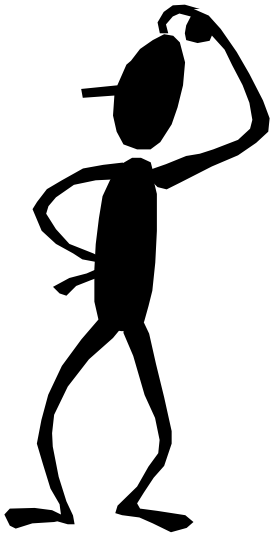
What is an operating system?

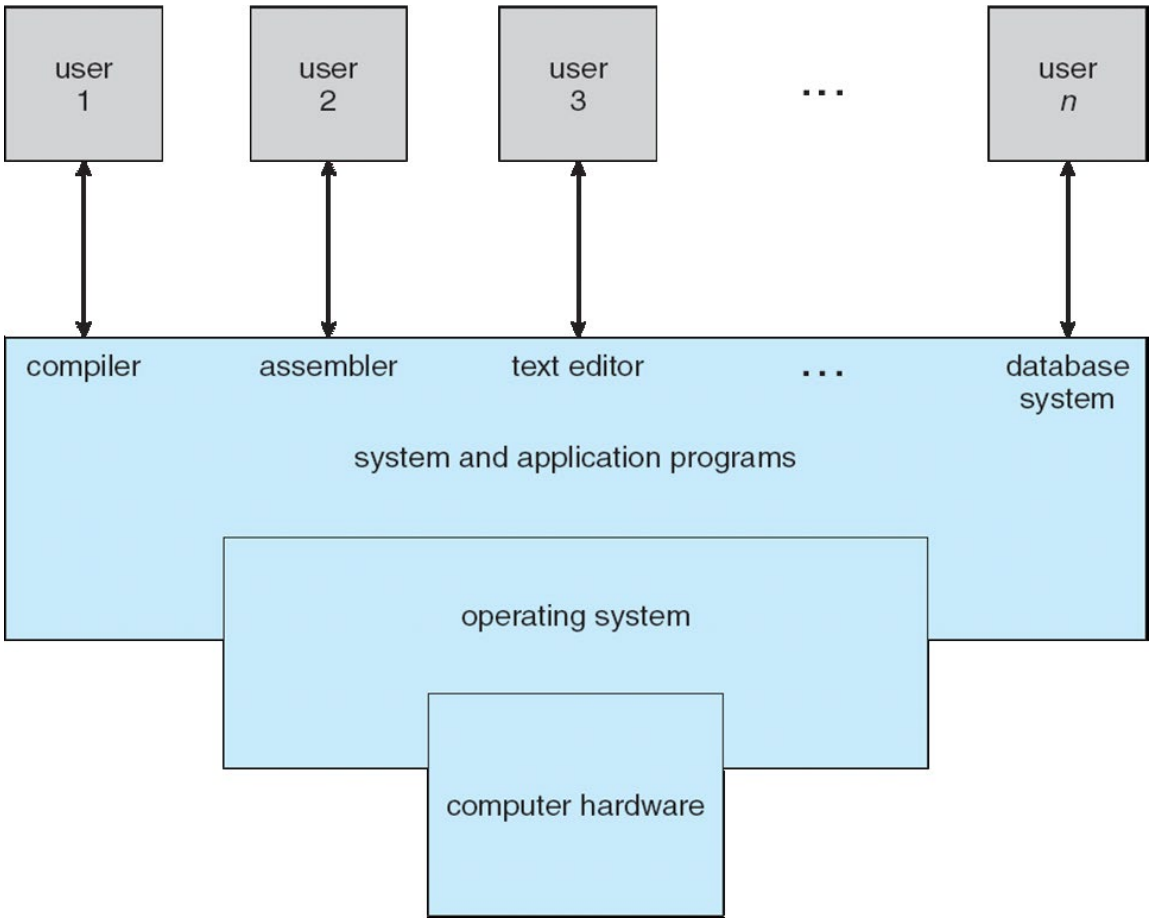
- What is an *Operating System*?
  - The software interface between hardware and its users
- Operating systems:
  - Execute user and system programs
  - Manage and coordinate computer hardware
  - Serve as resource allocators
  - Are typically interrupt-driven



Computer System Structure

- Computer system can be divided into four components:
  - Hardware – provides basic computing resources
    - CPU, memory, I/O devices
  - Operating system
    - Controls and coordinates use of hardware among various applications and users
  - Application programs – define the ways in which the system resources are used to solve the computing problems of the users
    - Word processors, compilers, web browsers, database systems, video games
  - Users
    - People, machines, other computers

Four Components of a Computer System



What Operating Systems Do

- Depends on the point of view
- Users want convenience, **ease of use**
  - Don't care about **resource utilization**
- But shared computer such as **mainframe** or **minicomputer** must keep all users happy
- Users of dedicate systems such as **workstations** have dedicated resources but frequently use shared resources from **servers**
- Handheld computers are resource poor, optimized for usability and battery life
- Some computers have little or no user interface, such as embedded computers in devices and automobiles

Operating System Definition

- OS is a **resource allocator**
  - Manages all resources
  - Decides between conflicting requests for efficient and fair resource use
- OS is a **control program**
  - Controls execution of programs to prevent errors and improper use of the computer

Operating System Definition (Cont.)

- No universally accepted definition
- "Everything a vendor ships when you order an operating system" is good approximation
  - But varies wildly
- "The one program running at all times on the computer" is the **kernel**. Everything else is either a system program (ships with the operating system) or an application program.

System software

- What is system software?
  - Computer programs that directly *control* the operations of the computer and its devices
- Operating systems:
  - Coordinate and orchestrate all activities of the hardware devices in a computer
  - Provide both a *Graphical User Interface* (*GUI*) and a *Command-Line Interface* (*CLI*) for its users

Operating system design goals

- From a user's perspective:
  - easy to use
  - easy to learn
  - reliable
  - safe
  - fast
  - etc.

- System goals:
  - reliability
  - flexibility
  - extensibility
  - speed(y)
  - efficiency
  - maintainability
  - etc.

Operating system services (i)

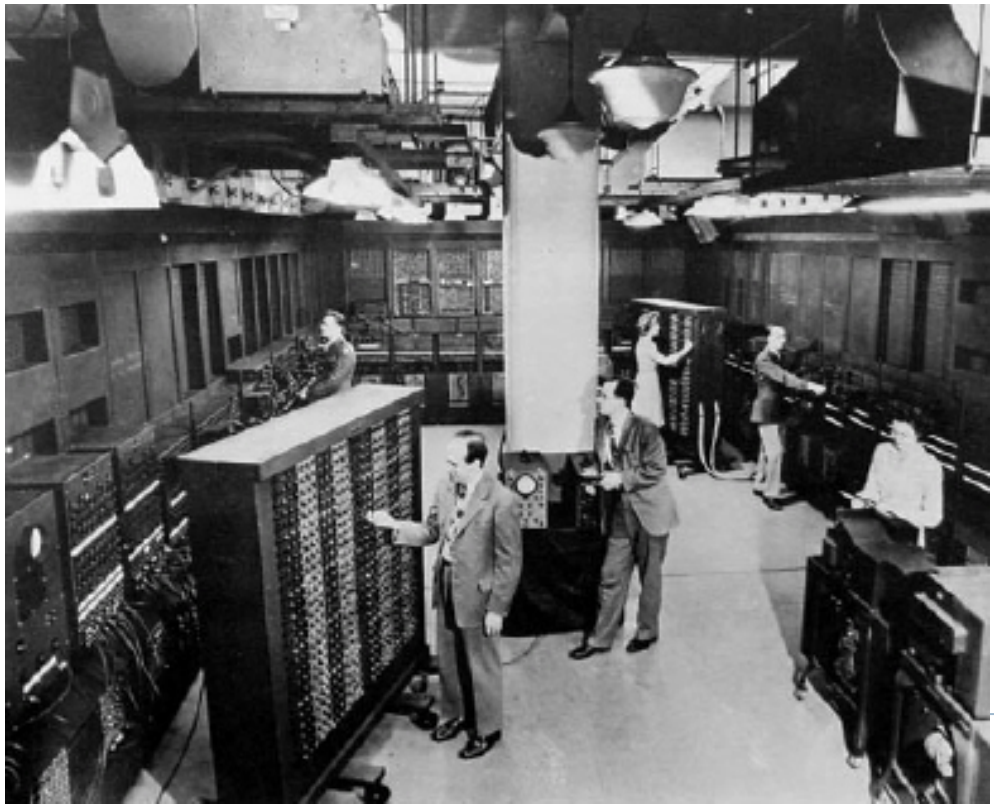
- An operating system provides *services*:
  - Program execution
    - Load programs into memory, run/suspend/halt programs, handle/display errors
  - I/O operations
    - Seamlessly interact with I/O devices, including disks, networks connection, etc.
  - Filesystem manipulation
    - Read/write/traverse filesystem directories, read/write files, enforce permissions, search for files

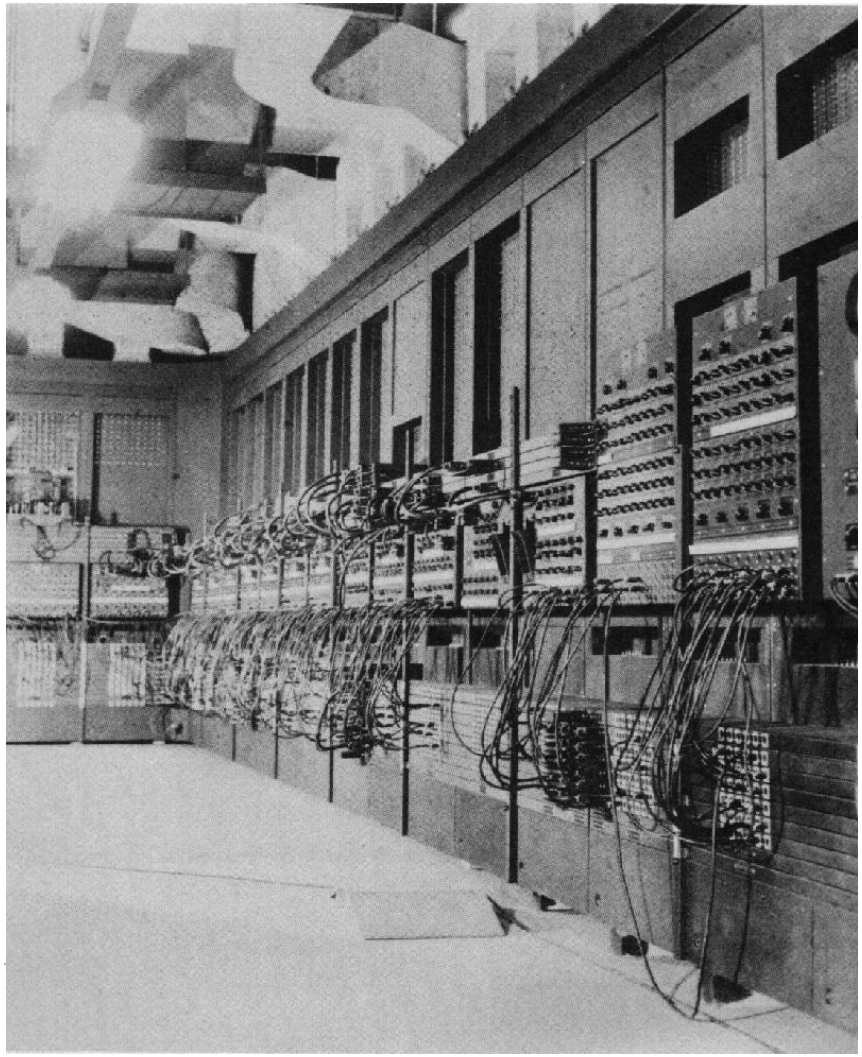
Operating system services (ii)

- Other operating system *services*:
  - Inter-Process Communications (IPC)
    - Processes exchange information via shared memory, message passing, sockets, pipes, files, etc.
    - Often spans multiple computers and networks
  - Error detection and recovery
    - Detect errors in CPU, memory, I/O devices, processes, network connections, etc.
    - Recover from errors gracefully, ensuring correct and consistent operations

History of operating systems (i)

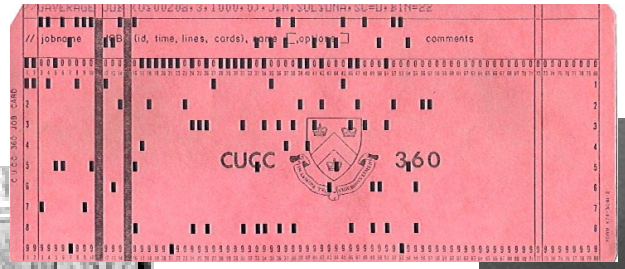
- In the beginning...
  ...the 1940s





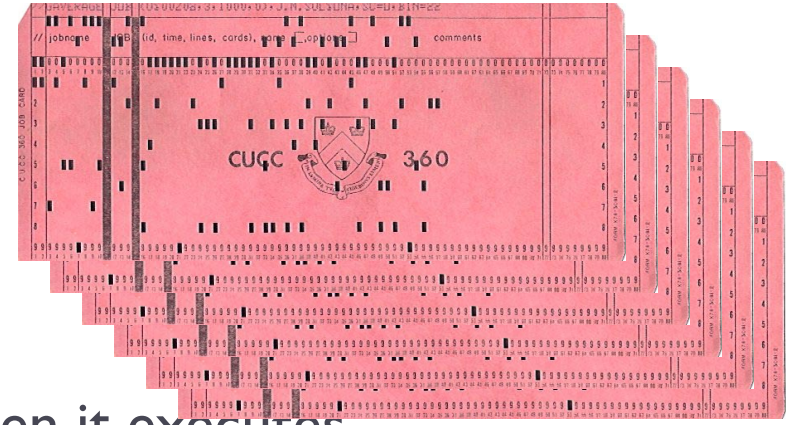
History of operating systems (ii)

- Automation in the 1950s with punch cards



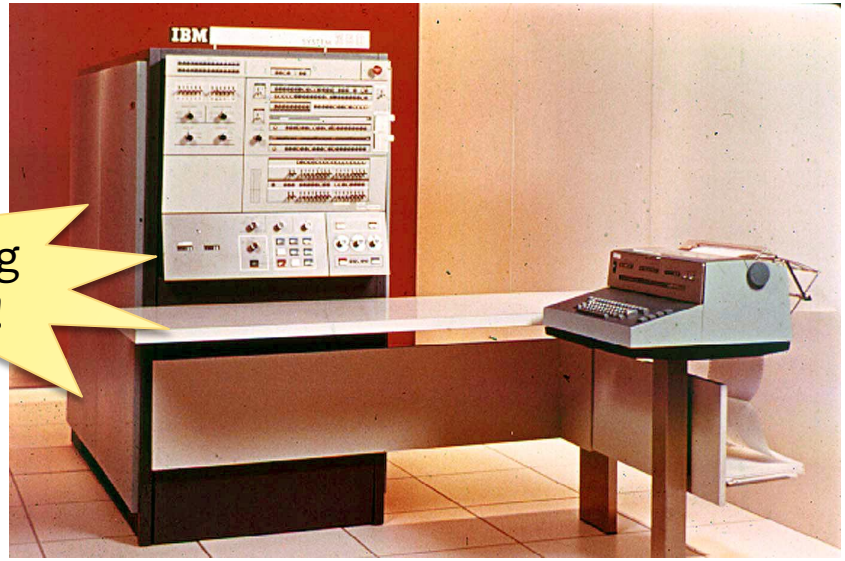
Batch jobs

- A *job* is a unit of work submitted by a user to the operating system
- Jobs typically consist of:
  - a program either in a source language or in "executable" binary form
  - input data used by the program when it executes

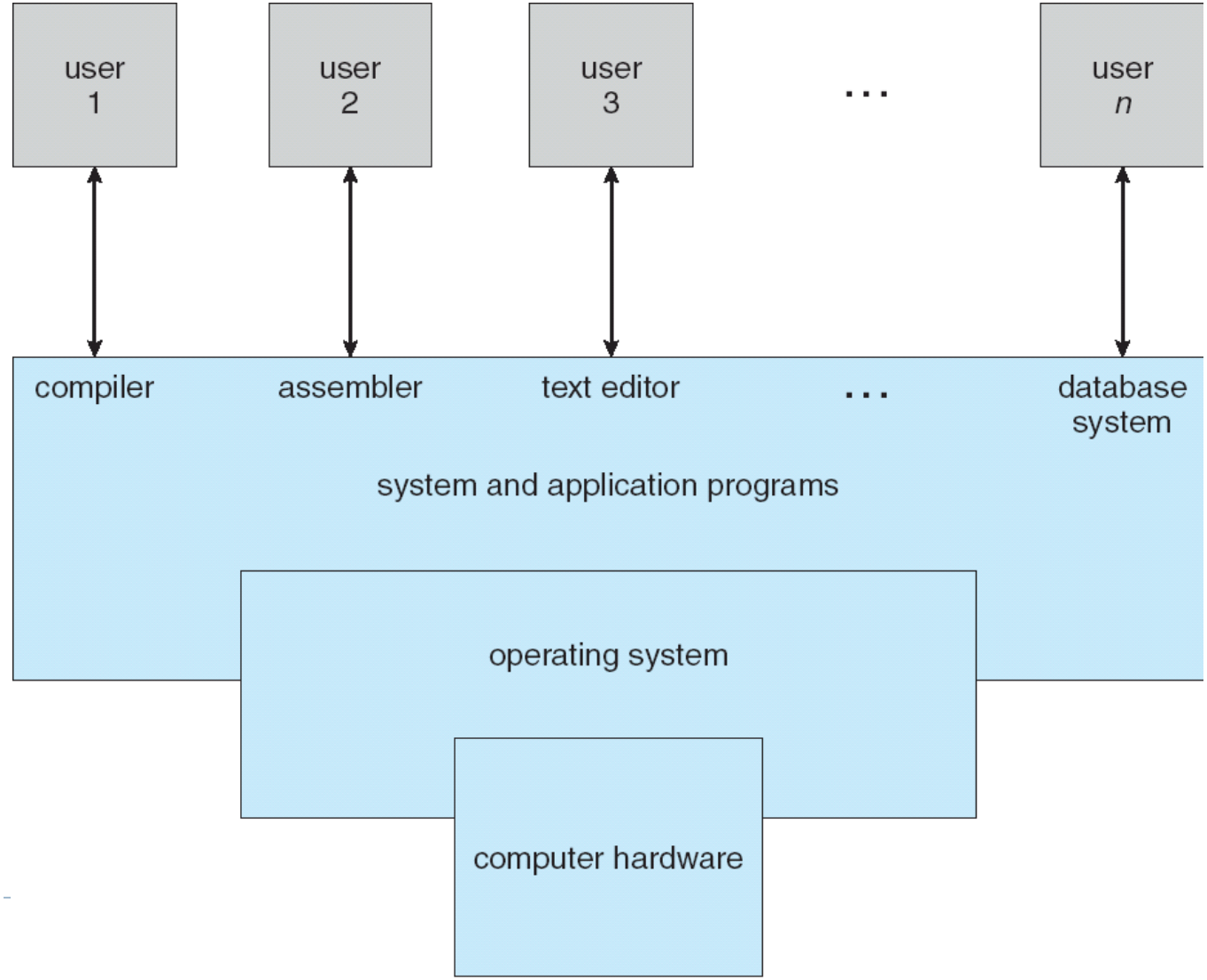


History of operating systems (iii)

- IBM 360 introduced (in 1964)

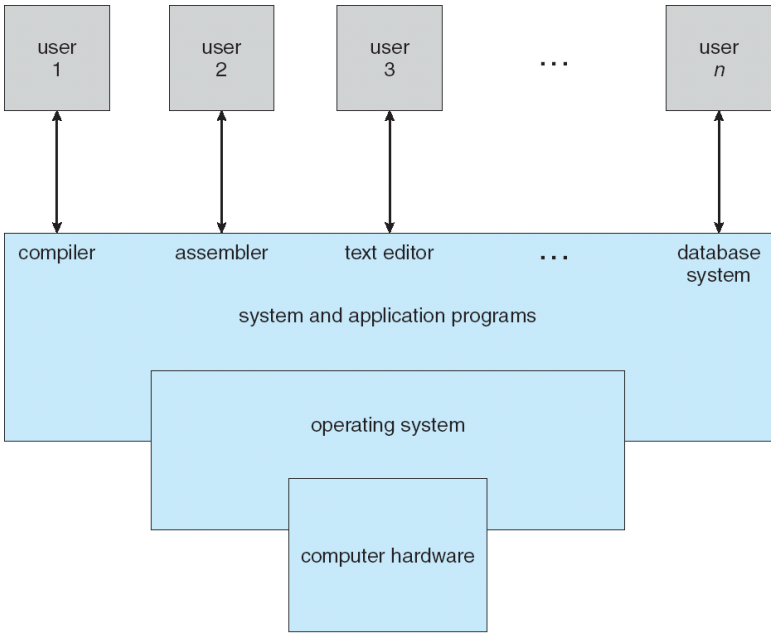


Multiprogramming (i)



Multiprogramming (ii)

- In multiprogramming, several jobs reside in memory simultaneously
  - CPU use is shared and managed by the operating system

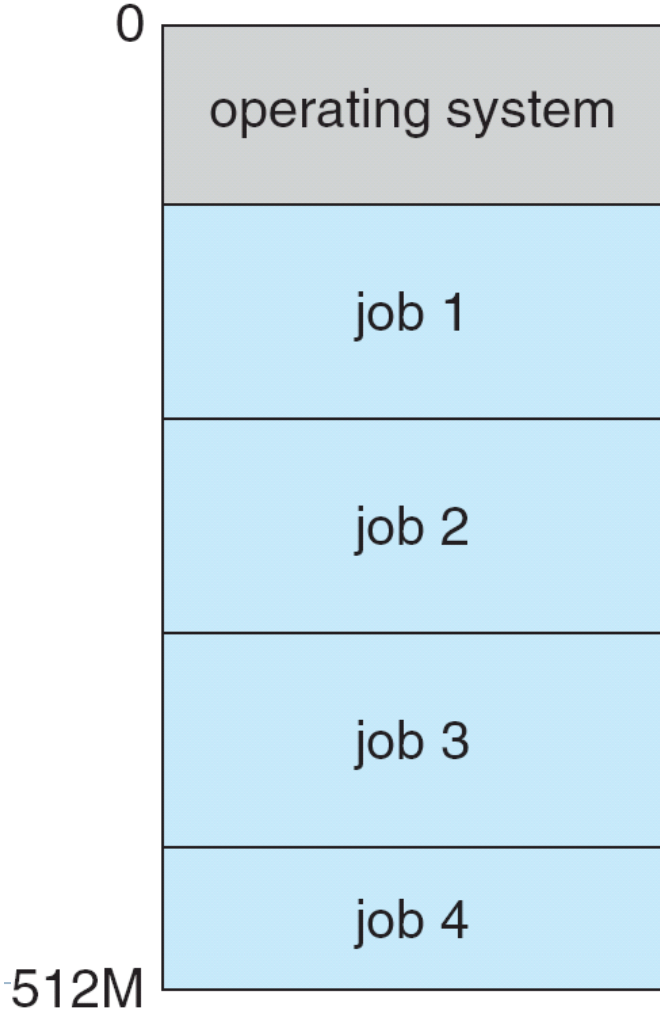


Multiprogramming (iii)

- Multiprogramming provides efficient use of the computer (CPU) and its resources (I/O)
  - One user cannot keep the CPU and I/O devices busy at all times
  - Multiprogramming organizes jobs such that the CPU always has exactly one to execute

Multiprogramming (iv)

- Computer is often idle – why?
  - CPU and hardware *significantly* faster than I/O
  - When a user or process is blocked waiting for I/O, the operating system switches to another job
  - A subset of jobs is stored in memory, awaiting CPU or I/O

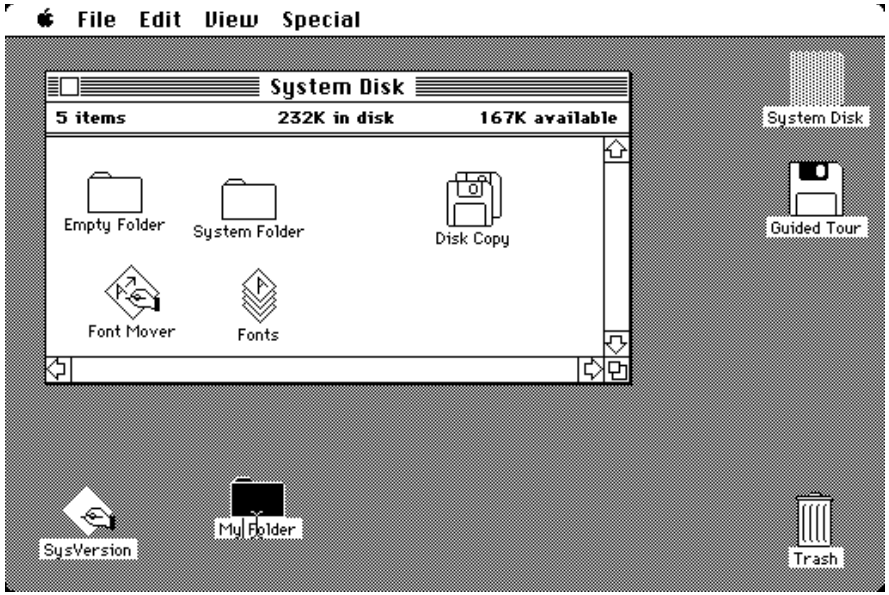


Timesharing and multitasking

- To ensure fairness, use a *timesharing scheme* in which the CPU cycles through all jobs
  - Each job is given a fixed amount of CPU time
  - Switching from one running job (or process) to another is called a *context switch*
  - A process may relinquish its time if blocked on an I/O request

History of operating systems (iv)

- Text CRTs (1970s) to an early Mac (1984)



History of operating systems (v)

- Personal computer revolution (1970s/80s)

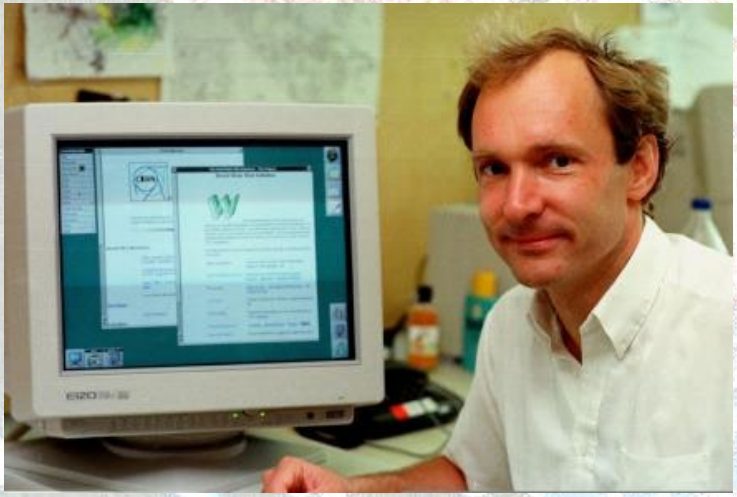
History of operating systems (vi)

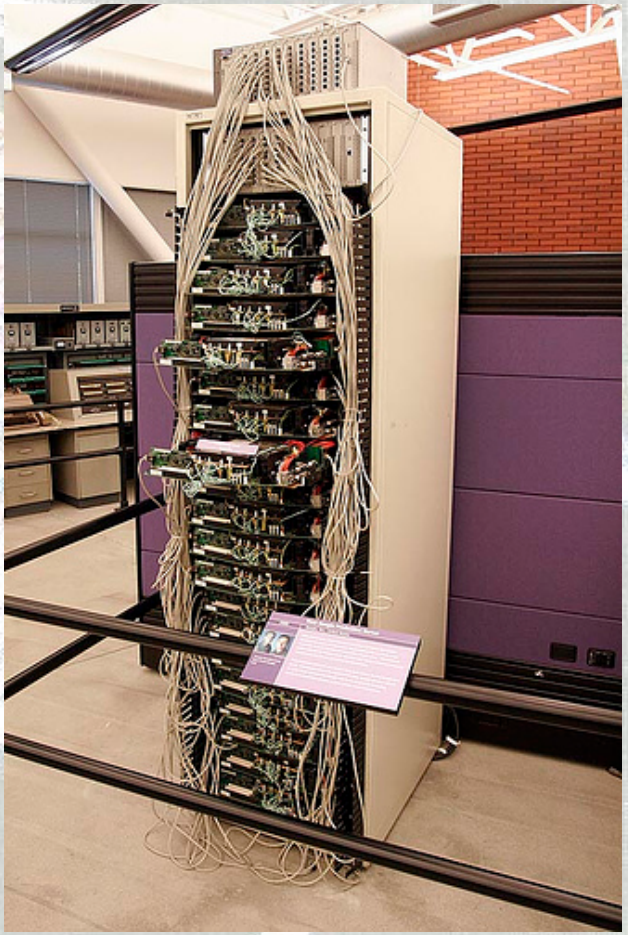
- The war begins...

sell your stock....

History of operating systems (vii)

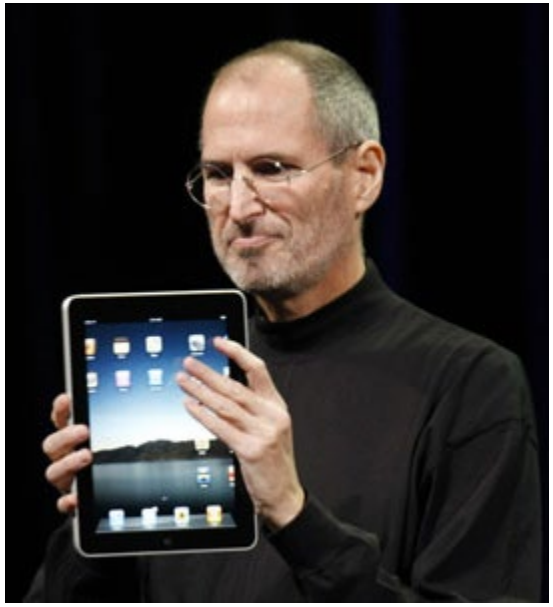
- World Wide Web and Internet revolution (1990s/2000s)





History of operating systems (viii)

- Mobile revolution (2010s)

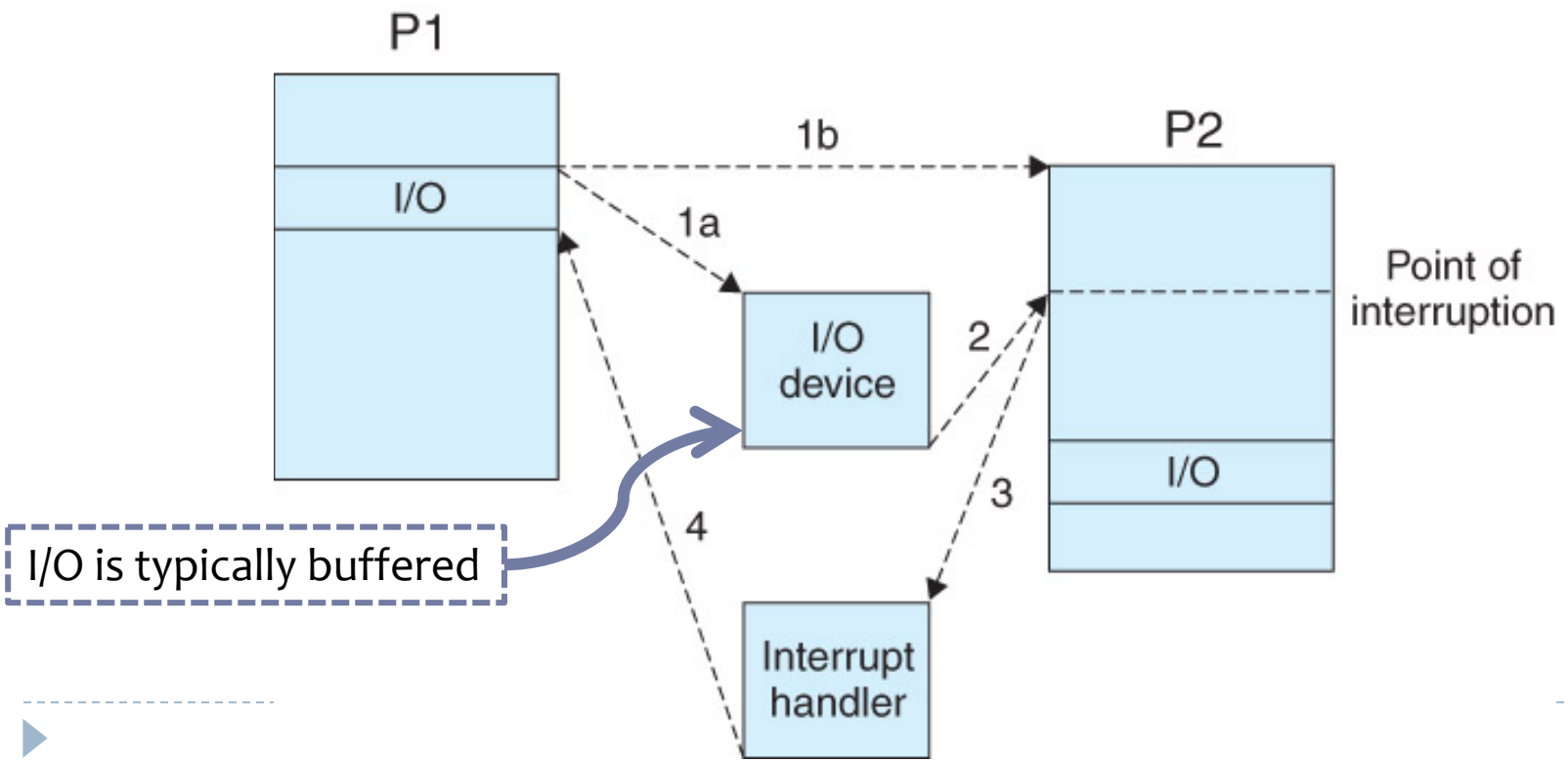


We interrupt this program…

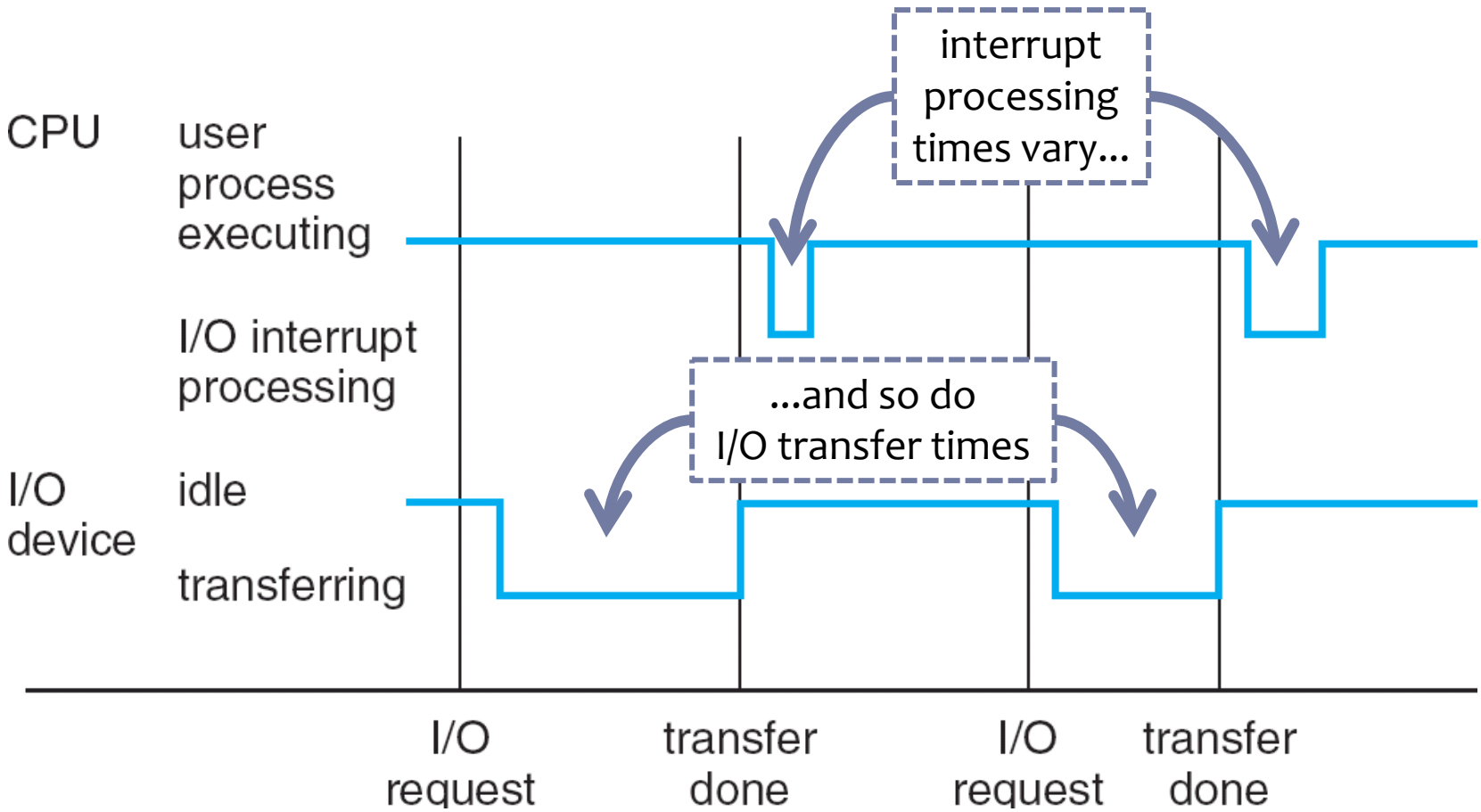
- Software instructions are executed by a Central Processing Unit (CPU)
  - An external hardware event triggers an interrupt by signaling the CPU
    - e.g. mouse movement, keyboard event
  - Software triggers an interrupt by executing a system call
    - e.g. disk read, graphics output, printer output

Interrupt mechanism

- Interrupts are handled much like calling a function in a programming language

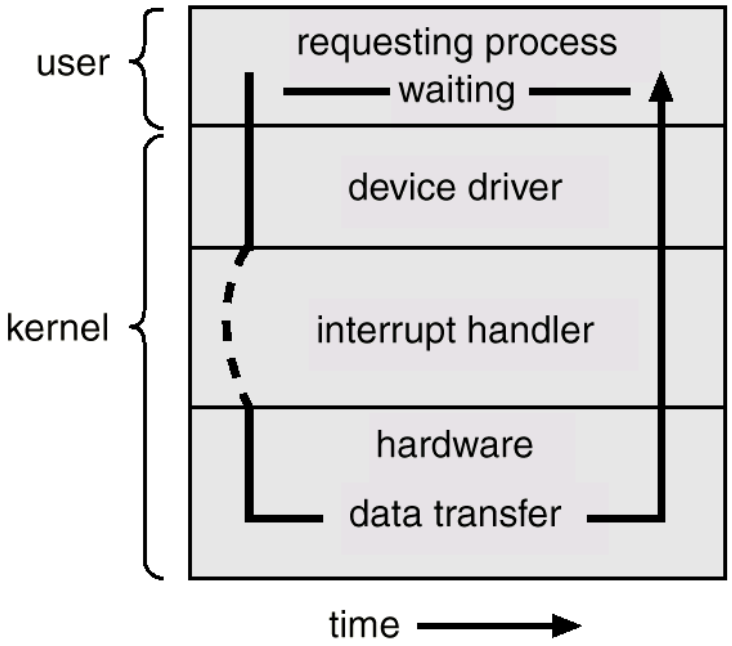


Typical interrupt timeline

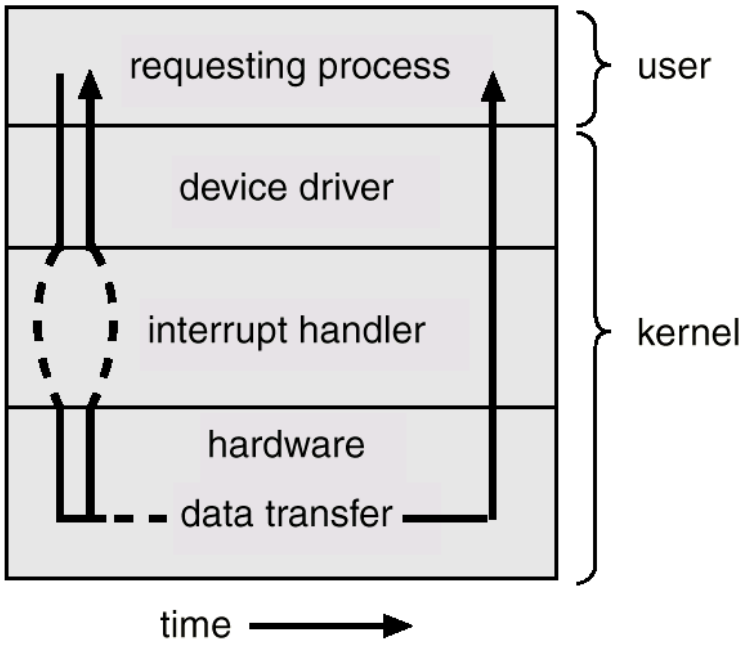


Synchronous and Asynchronous I/O

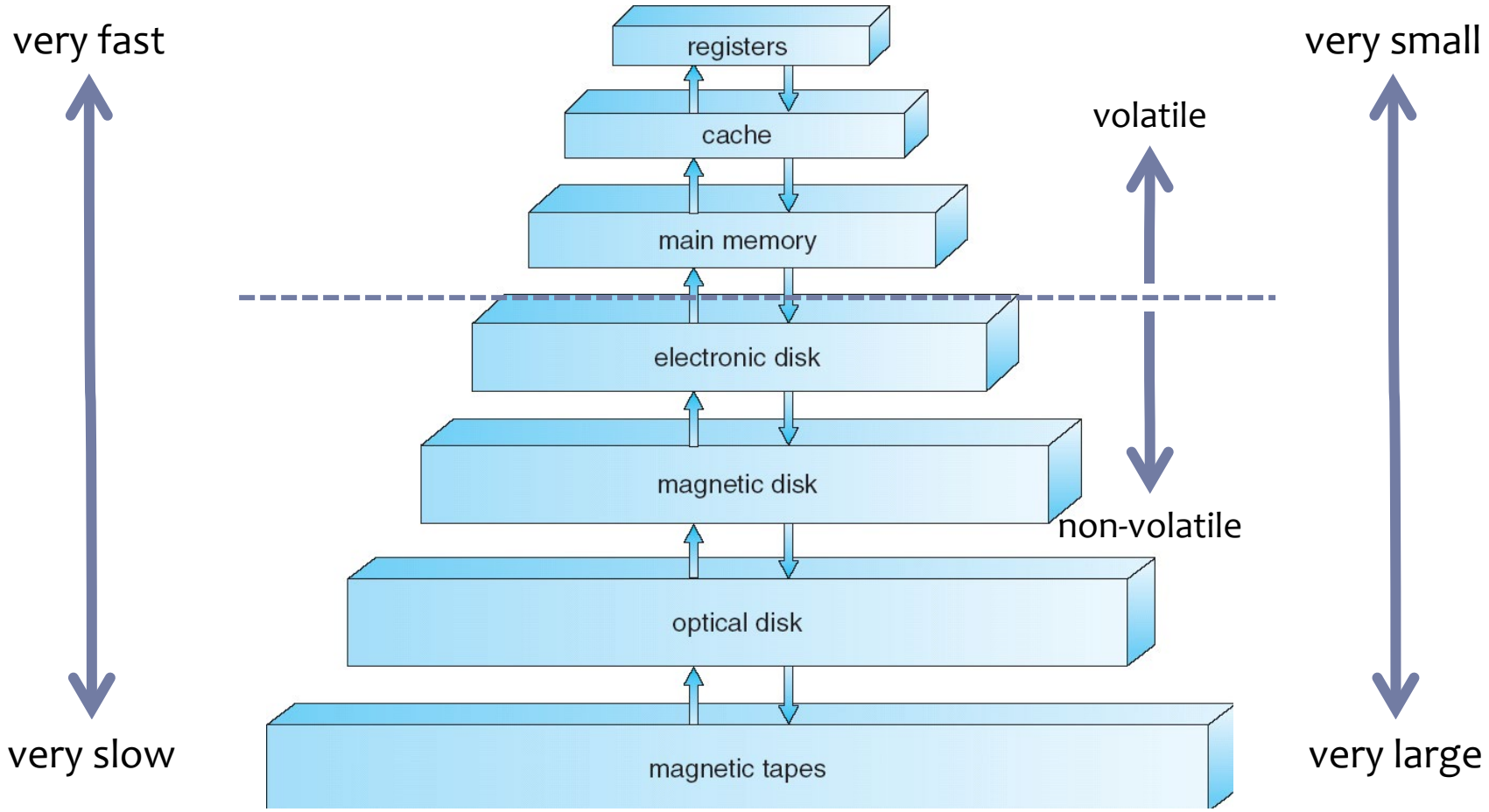
Synchronous



Asynchronous

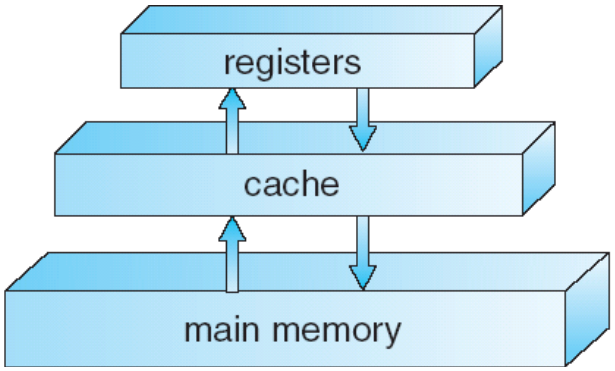


Hierarchical storage architecture



Caching (i)

- Caching is a technique in which data is temporarily stored in a smaller and faster memory component
  - Why implement caching in an operating



Caching (ii)

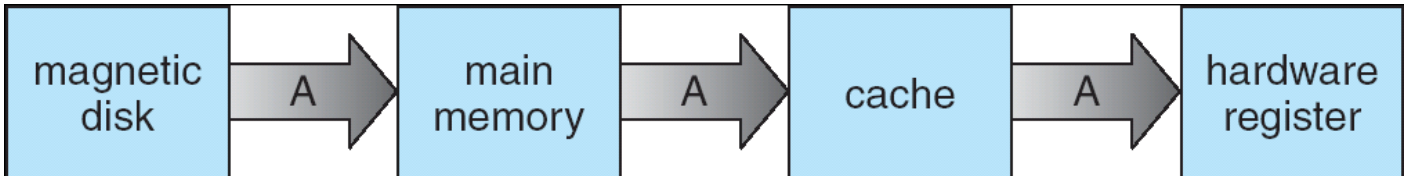
- A key goal in operating system design is achieving *fast and efficient performance*

Level	1	2	3	4
Name	registers	cache	main memory	disk storage
Typical size	< 1 KB	> 16 MB	> 16 GB	> 100 GB
Implementation technology	custom memory with multiple ports, CMOS	on-chip or off-chip CMOS SRAM	CMOS DRAM	magnetic disk
Access time (ns)	0.25 – 0.5	0.5 – 25	80 – 250	5,000.000
Bandwidth (MB/sec)	20,000 – 100,000	5000 – 10,000	1000 – 5000	20 – 150
Managed by	compiler	hardware	operating system	operating system
Backed by	cache	main memory	disk	CD or tape

Caching (iii)

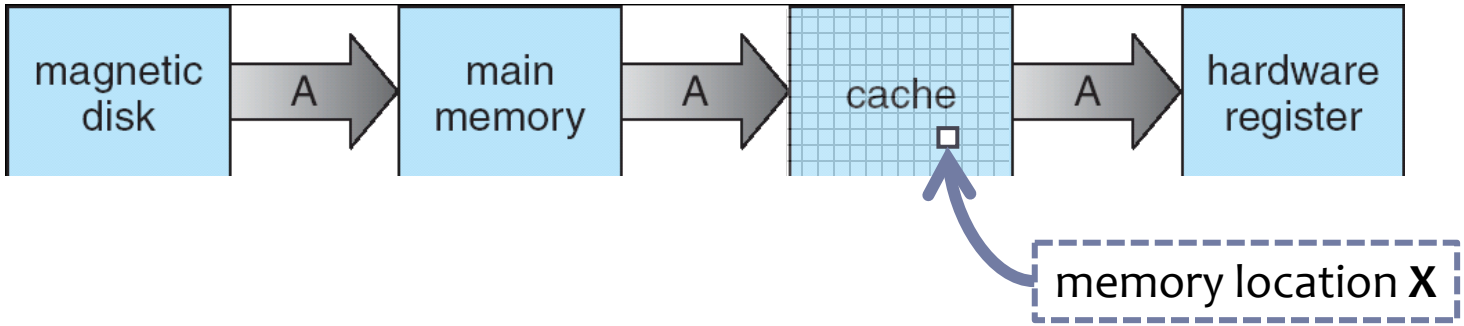
- What's the caching algorithm?
  - When the operating system attempts to read from memory, check to see if the requested data is already in the cache
  - If it is, data is read from the cache (*fast*!)
  - If not, data is copied from memory to the cache

(*maybe next time...*)



Principle of locality

- When a running program reads from memory location **X**, the *principle of locality* predicts that the next memory location requested will be near **X**



- Store *pages* of data in a cache, where each page is typically the same size (e.g. 64KB)